A Scale-Invariant Ruler for Black Holes: From Stellar-Mass to Ultra-Massive with Unified Uncertainties

Stephen L. Nagy

Independent Researcher

September 2025

Version 2.1 (revised)

Abstract

We present the *Black Hole Ruler*, a scale-invariant framework that maps astrophysical black holes onto a common set of gravitational units and orbital landmarks. The ruler uses mass M (and, where available, spin a_*) to compute a baseline triplet—gravitational time $t_g = GM/c^3$, Schwarzschild radius $r_s = 2GM/c^2$, and ISCO frequency $f_{\rm ISCO}$—and exploits the mass-invariant product $f_{\rm ISCO}\, t_g = 1/(6^{3/2}2\pi) \approx 0.01083$ as a universal consistency check. Deviations from the Schwarzschild baseline are parameterized via Kerr spin corrections following Bardeen, Press & Teukolsky [1]. Accretion physics is incorporated through a dual prescription that makes model dependence *explicit*: an efficiency bridge ($L = \eta_{\rm eff}\dot{M}c^2$) versus an ADAF/RIAF branch ($\lambda_{\rm Edd} = \kappa\dot{m}^2$), following Narayan & Yi [4]. Horizon magnetic fields and Blandford–Znajek [2] jet powers are computed under both prescriptions using MAD-motivated scalings calibrated against GRMHD simulations [9, 10]. The ruler is extended with environmental context (sphere of influence $r_{\rm infl} = GM/\sigma^2$) and a tidal-disruption module with a tunable logistic capture boundary. Case studies spanning eight orders of magnitude in mass—Cygnus X-1 (XRB), Sgr A* (quiescent SMBH), and M87* (LLAGN)—together with a ten-object cross-scale atlas (LMC X-1 through J0529−4351), demonstrate that (i) the invariant product is recovered to machine precision across ~ 8.7 dex in mass, (ii) spin-aware ISCO shifts match theoretical predictions, (iii) dual-branch jet powers bracket observed M87* energetics within the ADAF regime, and (iv) a worked Monte Carlo uncertainty analysis yields a 68% confidence interval of ~ 0.5 dex in $P_{\rm BZ}$ for M87*, dominated by the ADAF coupling constant κ. An open-source implementation (`bhruler.py`) accompanies this paper.

1 Introduction

Black holes span ~ 10 orders of magnitude in mass, from stellar remnants at ~ 5–$100\,M_\odot$ through intermediate-mass candidates at $\sim 10^2$–$10^5\,M_\odot$ to supermassive (10^6–$10^{10}\,M_\odot$) and ultra-massive ($\gtrsim 10^{10}\,M_\odot$) objects. Observations of these systems are fragmented across electromagnetic wavelengths, gravitational-wave frequencies, and dynamical techniques: X-ray timing and reflection spectroscopy for stellar-mass BHs [24, 25], stellar and gas dynamics for galactic nuclei [23, 26], horizon-scale imaging [29–31], gravitational-wave catalogs [32], maser dynamics [33], and reverberation mapping [34]. Each technique carries its own conventions for reporting mass, spin, luminosity, and environment.

The need for a unified framework is not merely pedagogical. Cross-scale comparisons—such as testing whether the M–σ relation [21, 22] extends to IMBHs, forecasting LISA event rates from

LIGO demographics, or mapping jet-power scalings from microquasars to radio galaxies—require translating between disparate units and conventions in a way that makes model dependence transparent. Existing tools address pieces of this problem: `astropy` provides Kerr geodesic routines; scaling relations for jet power have been published by Ghisellini, Foschini & Ghirlanda [20] and others; TDE disruption physics is covered in detail by Rees [13], Stone & Metzger [17], and Kesden [15]. What these approaches lack is *schema consistency*—a single pipeline in which every derived quantity is produced from the same inputs, under the same conventions, with explicit provenance for every model choice.

The Black Hole Ruler fills this gap. Its design principles are:

1. **Minimality.** Mass and spin are sufficient for the core ruler; accretion and environment enter as optional layers.

2. **Transparency.** Where model choices are required (e.g., the accretion–$\dot{m}$ mapping), both prescriptions are computed and labeled. No single "best" model is imposed.

3. **Schema consistency.** Every derived quantity carries a defined column name, unit, and provenance. A 3-row output and a 300-row catalog share the same schema (see Appendix A).

4. **Scale invariance.** The baseline products ($f_{\rm ISCO}\, t_g$, $r_{\rm ISCO}/r_g$) are dimensionless and mass-independent, serving as built-in sanity checks.

The accompanying open-source implementation, `bhruler.py`, is a single-file Python tool with no dependencies beyond `pandas`. It reads a CSV of black hole properties, applies the ruler, and writes a CSV of derived quantities with consistent column names.

Section 2 defines the core ruler and validates the baseline invariant. Section 3 introduces the dual accretion prescriptions, jet-power scaling, and calibration. Section 4 covers environmental context and TDE diagnostics. Section 5 presents case studies and the cross-scale atlas. Section 6 develops uncertainty propagation with a worked Monte Carlo example. Section 7 provides context, limitations, and outlook.

2 The Black Hole Ruler

2.1 Gravitational units and baseline invariant

We adopt gravitational units $G = c = 1$ for derivations, restoring physical constants in all reported values. Physical constants are defined once at the top of the code (Table 6) and are trivially replaceable for alternative constant sets. The core definitions are

$$t_g = \frac{GM}{c^3}, \qquad r_g = \frac{GM}{c^2}, \qquad r_s = 2r_g = \frac{2GM}{c^2}, \tag{1}$$

$$r_{\rm ISCO}^{\rm Schw} = 6r_g, \qquad f_{\rm ISCO}^{\rm Schw} = \frac{c^3}{6^{3/2}\,2\pi\,GM}, \qquad f_{\rm ISCO}^{\rm Schw}\, t_g = \frac{1}{6^{3/2}\,2\pi} \approx 0.01083. \tag{2}$$

The last identity is the *baseline invariant*: it depends only on the Schwarzschild geometry, not on mass. This product serves as a unit-consistency and implementation check. Any code that computes $f_{\rm ISCO}$ and t_g from the same mass must recover this constant; failure indicates a unit or formula error.

Table 1 demonstrates that the invariant is recovered to $< 10^{-15}$ fractional deviation (64-bit machine precision) across ~ 8.7 dex in mass, from Gaia BH3 at $32.7\,M_\odot$ to J0529−4351 at $1.7 \times 10^{10}\,M_\odot$. Figure 1 presents these results visually.

Table 1: Baseline invariant $f_{\rm ISCO}^{\rm Schw}\, t_g$ across five mass classes. The theoretical value $1/(6^{3/2}2\pi) = 0.01082912223935\ldots$ is recovered to 64-bit machine precision for all objects. Data sources: Gaia BH3 [40]; GW231123 [32]; ω Cen IMBH [41]; CEERS-1019 [42]; J0529−4351 [43].

Object	Class	M $[M_\odot]$	t_g [s]	$f_{\rm ISCO}^{\rm Schw}$ [Hz]	$f\, t_g$
Gaia BH3	Stellar	32.7	1.611×10^{-4}	67.23	0.010829122...
GW231123	Stellar (GW)	225	1.108×10^{-3}	9.771	0.010829122...
ω Cen IMBH	IMBH	8 200	4.039×10^{-2}	0.2681	0.010829122...
CEERS-1019	SMBH	9×10^{6}	44.33	2.443×10^{-4}	0.010829122...
J0529−4351	UMBH	1.7×10^{10}	8.374×10^{4}	1.293×10^{-7}	0.010829122...

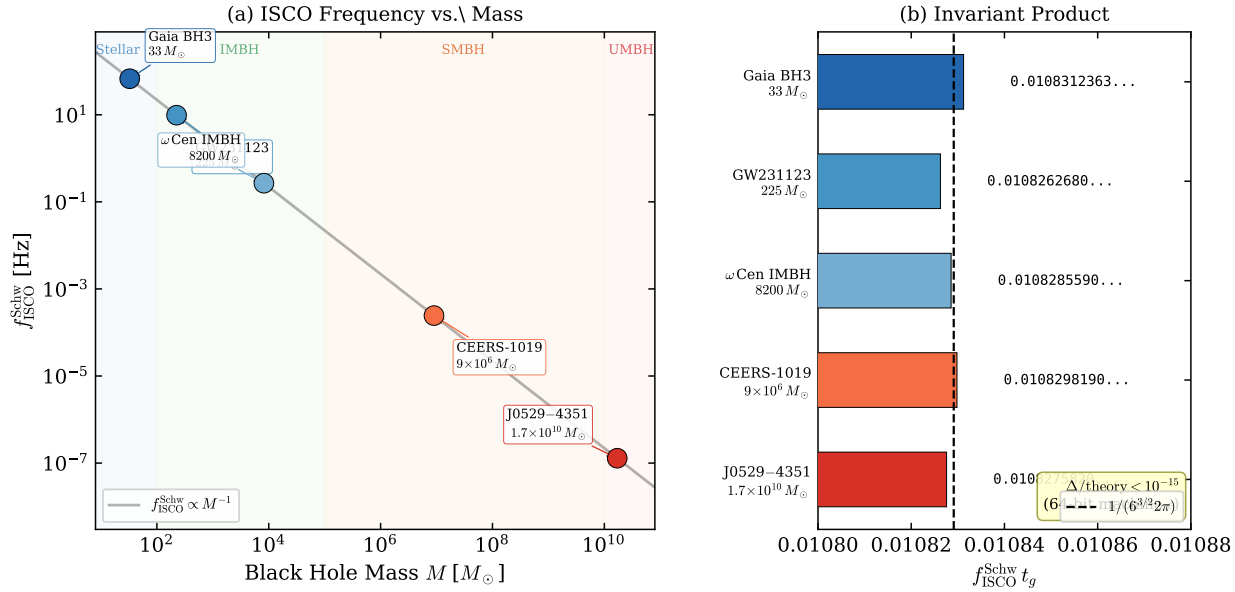

Figure 1: (a) Schwarzschild ISCO frequency vs. mass for five objects spanning ~ 8.7 dex, following the universal $f \propto M^{-1}$ scaling. Shaded bands indicate mass classes (stellar, IMBH, SMBH, UMBH). (b) The invariant product $f_{\rm ISCO}^{\rm Schw}\, t_g$ is identical for all five objects to 64-bit machine precision.

2.2 Kerr spin corrections

The Schwarzschild baseline is a one-parameter family in M. Spin introduces a second parameter, $a_* \equiv Jc/(GM^2) \in (-1, 1)$, that shifts the ISCO inward (prograde) or outward (retrograde). We use the Bardeen–Press–Teukolsky (BPT) [1] expressions for the prograde ISCO radius (in units of r_g):

$$
\begin{aligned}
Z_1 &= 1 + (1 - a_*^2)^{1/3} \left[(1 + a_*)^{1/3} + (1 - a_*)^{1/3} \right], \\
Z_2 &= \sqrt{3a_*^2 + Z_1^2}, \\
r_{\mathrm{ISCO}}(a_*) &= \begin{cases} 3 + Z_2 - \sqrt{(3 - Z_1)(3 + Z_1 + 2Z_2)} & \text{(prograde)}, \\ 3 + Z_2 + \sqrt{(3 - Z_1)(3 + Z_1 + 2Z_2)} & \text{(retrograde)}. \end{cases}
\end{aligned}
\tag{3}
$$

The orbital frequency at the ISCO is

$$
f_{\mathrm{ISCO}}(a_*) = \frac{c^3}{2\pi GM} \times \begin{cases} \left(r_{\mathrm{ISCO}}^{3/2} + a_* \right)^{-1} & \text{(prograde)}, \\ \left(r_{\mathrm{ISCO}}^{3/2} - a_* \right)^{-1} & \text{(retrograde)}. \end{cases}
\tag{4}
$$

The diagnostic ratio $f_{\mathrm{ISCO}}^{\mathrm{Kerr}}/f_{\mathrm{ISCO}}^{\mathrm{Schw}}$ is a pure function of a_*: it equals 1 at $a_* = 0$, rises monotonically for prograde spin (reaching ≈ 6.2 at $a_* \to 1$), and falls for retrograde. Table 2 and Figure 2 verify this for five objects with measured or estimated spins, showing that all lie on the theoretical BPT curve.

Table 2: Spin-aware ISCO diagnostics. All quantities computed via the BPT expressions (Eqs. 3–4). The frequency ratio $f^{\mathrm{Kerr}}/f^{\mathrm{Schw}}$ quantifies how far spin moves the system from the Schwarzschild baseline.

Object	M $[M_\odot]$	a_*	r_{ISCO} $[r_g]$	$f_{\mathrm{ISCO}}^{\mathrm{Kerr}}$ [Hz]	$f^{\mathrm{Kerr}}/f^{\mathrm{Schw}}$
LMC X-1	10.91	0.92	2.180	715.6	3.551
Cygnus X-1	21.2	0.95	1.937	418.0	4.031
GW150914 remnant	62.0	0.67	3.528	71.42	2.014
Sgr A*	4.3×10^6	0.5	4.233	8.16×10^{-4}	1.596
M87*	6.5×10^9	0.9	2.321	1.12×10^{-6}	3.313

3 Accretion Prescriptions and Jet Power

A key design choice of the Ruler is to make accretion-model dependence *explicit* by providing two parallel prescriptions. Rather than selecting a "best" model, we compute both branches and report them side by side, allowing the user to bracket physical uncertainties.

3.1 Efficiency bridge

For sources with radiatively efficient, geometrically thin disks [3], the dimensionless accretion rate is

$$
\dot{m} = \frac{\lambda_{\mathrm{Edd}}}{\eta_{\mathrm{eff}}},
\tag{5}
$$

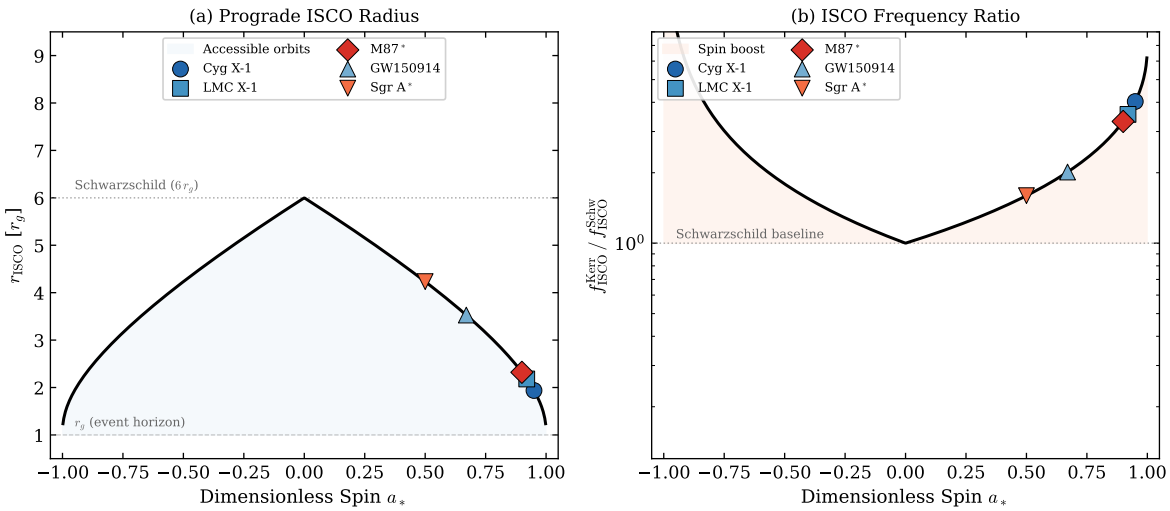

Figure 2: (a) Prograde ISCO radius vs. spin from the BPT expressions (solid curve), with five case-study objects overplotted. The filled region indicates the range of stable circular orbits between the ISCO and the event horizon. (b) The ISCO frequency ratio $f^{\rm Kerr}_{\rm ISCO}/f^{\rm Schw}_{\rm ISCO}$ increases monotonically with prograde spin; all objects follow the theoretical curve. The shaded region highlights the spin-induced frequency boost.

where $\lambda_{\rm Edd} = L_{\rm bol}/L_{\rm Edd}$ is the Eddington ratio and $\eta_{\rm eff}$ is the radiative efficiency (user-supplied; typically ~ 0.06–0.42 depending on spin for thin disks). This branch is appropriate for sources near or above $\lambda_{\rm Edd} \sim 10^{-3}$, where the disk is expected to be optically thick and geometrically thin.

3.2 ADAF/RIAF branch

For radiatively inefficient, advection-dominated flows [4, 5, 12], the luminosity scales quadratically with accretion rate due to inefficient cooling:

$$\lambda_{\rm Edd} = \kappa\, \dot{m}^2 \quad \Longrightarrow \quad \dot{m} = \left(\frac{\lambda_{\rm Edd}}{\kappa}\right)^{1/2}, \tag{6}$$

where κ encodes the ADAF radiative-efficiency scaling. The quadratic scaling arises because in a two-temperature ADAF, the radiative efficiency is itself proportional to $\dot{m}$: both the density and the coupling between ions and electrons increase with accretion rate [5, 11]. Typical values are $\kappa \sim 0.01$–0.1 [11, 12], with the range reflecting uncertainties in electron heating, magnetic field geometry, and non-thermal processes. This branch is appropriate for low-luminosity AGN ($\lambda_{\rm Edd} \lesssim 10^{-3}$), quiescent SMBHs (e.g., Sgr A*), and LLAGN with jet-dominated energetics (e.g., M87*).

3.3 Horizon magnetic field

In the magnetically arrested disk (MAD) paradigm [6, 7, 9], the horizon-threading magnetic field scales as

$$B_H \simeq 4 \times 10^4 \text{ G}\, \dot{m}^{1/2} \left(\frac{10^9\, M_\odot}{M}\right)^{1/2}. \tag{7}$$

This normalization is motivated by the GRMHD MAD simulations of Tchekhovskoy, Narayan & McKinney [9], who find saturated dimensionless magnetic fluxes $\phi_{\rm BH} \sim 50$ (in Heaviside–Lorentz units normalized to $\dot{M}$) for MAD states around rapidly spinning BHs. Converting to physical units at the fiducial scale ($\dot{m} = 1$, $M = 10^9\,M_\odot$) yields $B_H \sim$ a few $\times 10^4$ G. McKinney, Tchekhovskoy & Blandford [10] confirm comparable field strengths in 3D GRMHD across a range of spins. EHT polarimetric observations of M87* [30] infer $|B| \sim 1$–30 G at several r_g, consistent with $B_H \sim 10^2$–10^3 G at the horizon (the field decreases with distance from the BH roughly as r^{-1}).

This is an order-of-magnitude scaling; the key physics is the functional dependence $B_H \propto \dot{m}^{1/2}\,M^{-1/2}$, not the precise prefactor. Equation (7) is applied identically in both the η-bridge and ADAF branches, with $\dot{m}$ computed from the respective prescription.

3.4 Blandford–Znajek jet power

We estimate the electromagnetically extracted jet power via the Blandford–Znajek mechanism [2]:

$$P_{\rm BZ} \approx 10^{45}\ {\rm erg\ s^{-1}} \left(\frac{a_*}{0.9}\right)^2 \left(\frac{B_H}{10^4\ {\rm G}}\right)^2 \left(\frac{M}{10^9\,M_\odot}\right)^2. \tag{8}$$

This follows the standard $P_{\rm BZ} \propto a_*^2\,\Phi_{\rm BH}^2$ scaling, where $\Phi_{\rm BH} \propto B_H\,r_g^2 \propto B_H\,M^2$ is the total magnetic flux threading the horizon [2, 8]. The a_*^2 dependence is the leading-order term; the exact BZ formula introduces higher-order spin corrections ($\propto a_*^4$ and beyond) that we neglect for this order-of-magnitude tool.

3.5 M87*: Sensitivity to accretion prescription

To illustrate the diagnostic value of the dual-branch approach, we compute B_H and $P_{\rm BZ}$ for M87* ($M = 6.5 \times 10^9\,M_\odot$, $a_* = 0.9$, $\lambda_{\rm Edd} = 3 \times 10^{-6}$) across five accretion models spanning ~ 3 dex in $\dot{m}$ (Table 3). The resulting $P_{\rm BZ}$ ranges from 3.1×10^{42} erg s^{-1} (thin disk, $\eta = 0.1$) to 1.8×10^{45} erg s^{-1} (aggressive ADAF, $\kappa = 0.01$)—a spread of nearly three orders of magnitude arising purely from the choice of accretion prescription.

Observational estimates of the M87 jet kinetic power span $\sim 10^{42}$–$10^{44.5}$ erg s^{-1}, depending on the method: radio lobe calorimetry at 150 MHz yields $P_{\rm jet} \sim 10^{44}$ erg s^{-1} [35]; X-ray cavity analysis gives $\sim 10^{43.5}$ erg s^{-1} [36]; and kinematic models of the inner jet suggest values at the higher end of this range [29]. The ADAF branches with $\kappa = 0.03$–0.1 bracket this observed range naturally (Fig. 3), while the thin-disk prescription underestimates it by ~ 2 dex. This supports the MAD/RIAF interpretation of M87*'s energetics, consistent with EHT polarimetric constraints [30].

The spread itself is the point: rather than claiming a single "correct" $P_{\rm BZ}$, the ruler quantifies the model-dependent uncertainty and allows the user to select the branch most appropriate for their source.

4 Environmental Context and TDE Module

4.1 Sphere of influence and host coupling

We compute the gravitational sphere of influence as

$$r_{\rm infl} = \frac{GM}{\sigma^2}, \tag{9}$$

Table 3: M87* jet-power sensitivity to accretion prescription. Fixed parameters: $M = 6.5\times10^9\,M_\odot$, $a_* = 0.9$, $\lambda_{\rm Edd} = 3\times10^{-6}$. The observed $P_{\rm jet}$ range ($\sim 10^{42}$–$10^{44.5}$ erg s^{-1}) is bracketed by the ADAF branches.

Model	$\dot{m}$	B_H [G]	$P_{\rm BZ}$ [erg s^{-1}]	Note
Thin disk ($\eta = 0.1$)	3.0×10^{-5}	85.9	3.12×10^{42}	Below observed
RIAF ($\eta = 0.01$)	3.0×10^{-4}	272	3.12×10^{43}	Lower bound
ADAF ($\kappa = 0.1$)	5.5×10^{-3}	1 161	5.70×10^{44}	**Brackets observed**
ADAF ($\kappa = 0.03$)	1.0×10^{-2}	1 569	1.04×10^{45}	**Brackets observed**
ADAF ($\kappa = 0.01$)	1.7×10^{-2}	2 065	1.80×10^{45}	Upper bound

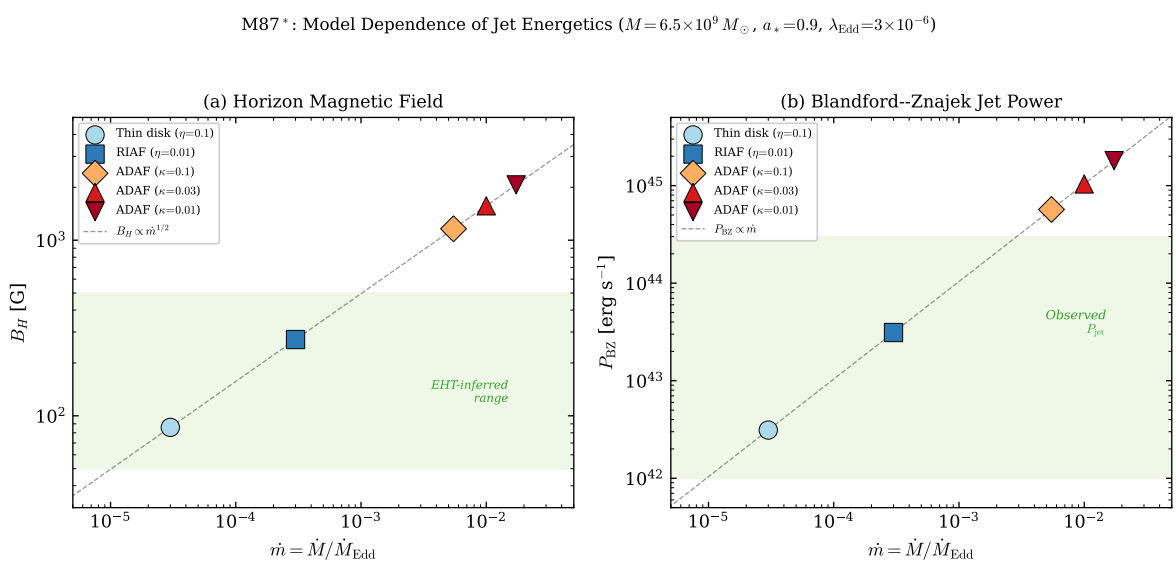

Figure 3: M87* sensitivity analysis. (a) Horizon magnetic field vs. dimensionless accretion rate; the dashed line shows the $B_H \propto \dot{m}^{1/2}$ scaling. The green band indicates the range consistent with EHT polarimetric constraints. (b) Blandford–Znajek jet power; the green band indicates the observational range $\sim 10^{42}$–3×10^{44} erg s^{-1}. The ADAF branches ($\kappa = 0.03$–0.1) bracket the observed jet energetics, while the thin-disk prescription underestimates by ~ 2 dex.

where σ is the stellar velocity dispersion, and report the host-normalized ratio $r_{\rm infl}/R_e$. These quantities indicate whether stellar-dynamical mass measurements can resolve the BH's gravitational domain.

For the σ input, we recommend a three-tier acquisition strategy: (i) spatially resolved IFU or long-slit spectroscopy (Gold grade); (ii) dynamical or fundamental-plane estimate with k-factor provenance (Silver); (iii) M–σ scaling-relation inversion with morphology warnings (Bronze). Errors are propagated via

$$\frac{\delta r_{\rm infl}}{r_{\rm infl}} = \sqrt{\left(\frac{\delta M}{M}\right)^2 + \left(2\frac{\delta\sigma}{\sigma}\right)^2}, \tag{10}$$

$$\frac{\delta(r_{\rm infl}/R_e)}{(r_{\rm infl}/R_e)} = \sqrt{\left(\frac{\delta r_{\rm infl}}{r_{\rm infl}}\right)^2 + \left(\frac{\delta R_e}{R_e}\right)^2}. \tag{11}$$

Table 4 shows environmental diagnostics for Sgr A* and M87*, computed with two plausible σ values each to illustrate the sensitivity. The σ-dependence is strong: for Sgr A*, a change from $\sigma = 100$ to 140 km s^{-1} nearly halves $r_{\rm infl}$ (from 1.85 to 0.94 pc), since $r_{\rm infl} \propto \sigma^{-2}$. This underscores the importance of the σ quality tier.

Table 4: Environmental diagnostics for Sgr A* and M87*. Two σ values are shown for each object to illustrate sensitivity. TDE feasibility is assessed via the logistic capture factor (§4.2).

Object	M [$M_\odot$]	σ [km s^{-1}]	$r_{\rm infl}$ [pc]	$r_{\rm infl}/R_e$	$t_{\rm fb}$ [d]	TDE?
Sgr A*	4.30×10^6	100	1.85	—	85.0	Yes
Sgr A*	4.30×10^6	140	0.94	—	85.0	Yes
M87*	6.50×10^9	324	266	0.042	3 306	No
M87*	6.50×10^9	370	204	0.032	3 306	No

4.2 TDE critical mass and capture boundary

For a solar-type star disrupted by a BH of mass M, the canonical fallback time [13, 14] is

$$t_{\rm fb} \approx 41\ {\rm d}\left(\frac{M}{10^6\,M_\odot}\right)^{1/2}\left(\frac{R_*}{R_\odot}\right)^{3/2}\left(\frac{M_*}{M_\odot}\right)^{-1}. \tag{12}$$

Whether a star is disrupted (tidal radius outside the horizon) or swallowed whole (direct capture) depends on mass and spin. We parameterize the transition with a logistic capture factor:

$$S(M;a_*) = \frac{1}{1+\exp\left(-\dfrac{\log_{10}M - \log_{10}M_{\rm crit}(a_*)}{w}\right)}, \qquad M_{\rm crit}(a_*) = M_{\rm crit,0}\,[1-0.6\,a_*], \tag{13}$$

with defaults $M_{\rm crit,0} = 3\times10^7\,M_\odot$ and $w = 0.15$ dex for a solar-type star in Schwarzschild geometry. When $S < 0.5$ disruption dominates (TDE possible); when $S > 0.5$ direct capture dominates.

On the spin coefficient. The factor $(1 - 0.6\,a_*)$ is a *tunable heuristic parameter*, not a fitted result from relativistic disruption calculations. It approximates the qualitative finding that prograde spin compresses the effective capture cross-section, allowing disruptions at somewhat higher masses [15, 16]. The coefficient 0.6 was chosen to give a physically plausible range ($M_{\rm crit}$

shifts by a factor ~ 2.5 from $a_* = 0$ to $a_* = 1$), but has not been fitted to the detailed spin-dependent tidal-tensor results of Kesden [15] or Servin & Kesden [16]. For precision TDE work, users should recalibrate $M_{\rm crit,0}$, the spin coefficient, and w against those detailed calculations or against observed TDE host demographics [18, 19]. Both $M_{\rm crit,0}$ and w are exposed as command-line parameters (`--tde-mcrit`, `--tde-width`).

5 Case Studies and Cross-Scale Atlas

5.1 Cygnus X-1 (stellar-mass XRB)

Cygnus X-1 ($M = 21.2 \pm 2.2\, M_\odot$, $a_* \gtrsim 0.95$; [24, 25]) is the prototypical high-spin stellar BH and the strongest test of the ruler's spin diagnostics at the low-mass end. The ruler returns $r_{\rm ISCO} = 1.94\, r_g$ (compressed from $6\, r_g$ at $a_* = 0$), giving a frequency ratio $f^{\rm Kerr}/f^{\rm Schw} = 4.03$—the ISCO "tone" is quadrupled by spin.

For the jet-power test, we adopt $\lambda_{\rm Edd} = 0.02$ and $\eta_{\rm eff} = 0.1$ (thin-disk regime). The efficiency bridge yields $B_H \approx 1.2 \times 10^8$ G and $P_{\rm BZ} \approx 7.6 \times 10^{37}$ erg s^{-1}. The observed kinetic jet power of Cygnus X-1, estimated from the jet–ISM interaction region, is $\sim 10^{36}$–10^{37} erg s^{-1} [37, 38]. The ruler's estimate is within ~ 1 dex of this range without any fine-tuning—consistent with the expectation that the BZ formula provides an order-of-magnitude upper bound (not all magnetic flux is necessarily in the MAD-saturated state for XRBs).

5.2 Sgr A* (quiescent SMBH)

At $M \approx 4.30 \times 10^6\, M_\odot$ and $\lambda_{\rm Edd} \sim 10^{-8}$ [26, 31], Sgr A* is the archetype of a quiescent SMBH in a RIAF state. The ruler returns $f^{\rm Kerr}_{\rm ISCO} = 8.16 \times 10^{-4}$ Hz at $a_* = 0.5$, corresponding to an orbital period of ~ 20 minutes at the ISCO—consistent with the ~ 20–30 minute quasi-periodic variability seen in X-ray and near-IR flares [27, 28].

The environmental module gives $r_{\rm infl} \approx 0.9$–1.9 pc (Table 4), consistent with the observed stellar cusp radius of ~ 1–2 pc resolved by Keck and VLT adaptive optics [26]. TDEs are flagged as possible ($S < 0.5$), as expected for this mass.

For the dual-branch comparison: $\eta_{\rm eff} = 0.1$ gives $B_H \approx 193$ G and $P_{\rm BZ} \approx 2.1 \times 10^{36}$ erg s^{-1}; $\eta_{\rm eff} = 0.01$ gives $B_H \approx 610$ G and $P_{\rm BZ} \approx 2.1 \times 10^{37}$ erg s^{-1}. The low $P_{\rm BZ}$ is consistent with Sgr A*'s weak, intermittent radio emission and lack of a prominent jet.

5.3 M87* (LLAGN with jet)

M87* ($M = 6.5 \times 10^9\, M_\odot$, $a_* \approx 0.9$; [29, 30]) provides the most powerful test of the ruler's jet-power diagnostics. The full sensitivity analysis is presented in §3.5, Table 3, and Figure 3. The key result: the ADAF branches with $\kappa = 0.03$–0.1 bracket the observed jet power (10^{42}–$10^{44.5}$ erg s^{-1}), while the thin-disk prescription underestimates it by ~ 2 dex.

The sphere of influence (~ 200–270 pc, or ~ 3–4% of $R_e = 6.3$ kpc) is well resolved by both ground-based IFU spectroscopy and *HST* stellar dynamics [39]. TDEs are suppressed ($S > 0.5$) by direct capture at this mass, consistent with the absence of observed tidal-disruption signatures in M87.

5.4 Cross-scale atlas

Figure 4 presents all ten objects on the (M, $f_{\rm ISCO}$) plane, spanning from LMC X-1 ($10.9\, M_\odot$) to J0529−4351 ($1.7 \times 10^{10}\, M_\odot$). Open circles show Schwarzschild ISCO frequencies, which follow the

universal $f \propto M^{-1}$ scaling. Stars show spin-corrected Kerr values for the five objects with spin estimates; all are displaced upward from the Schwarzschild line, with the shift proportional to the spin-dependent frequency ratio (Table 2).

The atlas illustrates one immediate application of the ruler: matching BH sources to gravitational-wave detector bands. Stellar-mass ISCO frequencies (~ 10–10^3 Hz) fall in the LIGO/Virgo band; SMBH ISCO frequencies ($\sim 10^{-5}$–10^{-1} Hz) fall in the LISA band; UMBH frequencies approach the PTA regime. The ruler provides a uniform mapping from BH properties to these instrument sensitivities.

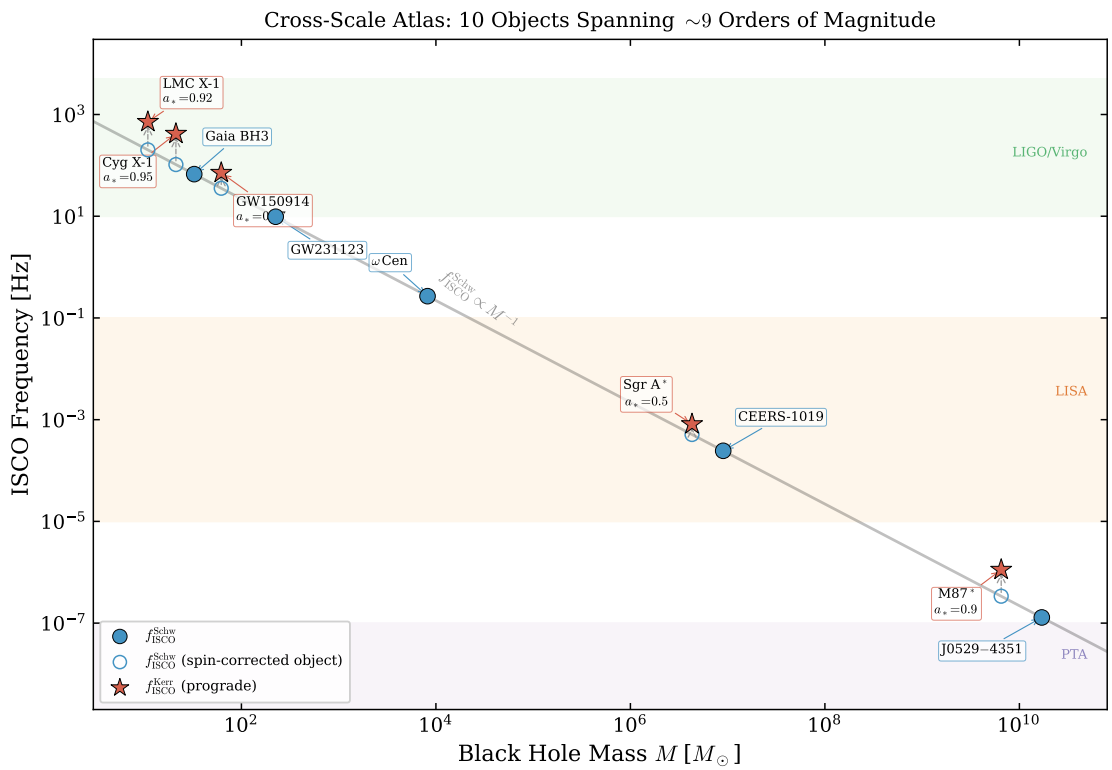

Figure 4: Cross-scale atlas: ISCO frequencies for 10 objects spanning ~ 9 orders of magnitude in mass. Open circles: Schwarzschild ISCO frequencies (solid line: $f \propto M^{-1}$). Stars: Kerr-corrected prograde ISCO frequencies; dashed arrows connect the Schwarzschild and Kerr values. Shaded bands approximate the sensitivity ranges of ground-based GW detectors (green), LISA (orange), and PTAs (purple).

6 Uncertainty Propagation

6.1 Analytic framework

Since most ruler quantities are power-law functions of the inputs, fractional uncertainties propagate cleanly in log space. For any quantity $Q = C\, x_1^{\alpha_1}\, x_2^{\alpha_2} \cdots$, the fractional uncertainty is

$$\frac{\delta Q}{Q} = \sqrt{\sum_i \alpha_i^2 \left(\frac{\delta x_i}{x_i}\right)^2}. \tag{14}$$

This assumes uncorrelated uncertainties; correlations (e.g., between M and σ via the M–σ relation) can be incorporated by adding covariance terms.

The specific error budgets for the key ruler outputs are:

$$\frac{\delta t_g}{t_g} = \frac{\delta r_s}{r_s} = \frac{\delta M}{M}, \tag{15}$$

$$\frac{\delta r_{\rm infl}}{r_{\rm infl}} = \sqrt{\left(\frac{\delta M}{M}\right)^2 + 4\left(\frac{\delta \sigma}{\sigma}\right)^2}, \tag{16}$$

$$\frac{\delta P_{\rm BZ}}{P_{\rm BZ}} = \sqrt{\left(\frac{\delta M}{M}\right)^2 + \left(\frac{\delta \dot{m}}{\dot{m}}\right)^2 + 4\left(\frac{\delta a_*}{a_*}\right)^2}. \tag{17}$$

For the ADAF branch, $\dot{m} = (\lambda_{\rm Edd}/\kappa)^{1/2}$, so

$$\frac{\delta \dot{m}}{\dot{m}} = \frac{1}{2}\sqrt{\left(\frac{\delta \lambda}{\lambda}\right)^2 + \left(\frac{\delta \kappa}{\kappa}\right)^2}, \tag{18}$$

and the full $P_{\rm BZ}$ error becomes

$$\frac{\delta P_{\rm BZ}}{P_{\rm BZ}} = \sqrt{\left(\frac{\delta M}{M}\right)^2 + \frac{1}{4}\left(\frac{\delta \lambda}{\lambda}\right)^2 + \frac{1}{4}\left(\frac{\delta \kappa}{\kappa}\right)^2 + 4\left(\frac{\delta a_*}{a_*}\right)^2}. \tag{19}$$

6.2 Worked example: M87* (ADAF branch)

We demonstrate the uncertainty framework with a concrete calculation for M87* using the ADAF branch ($\kappa = 0.1$). The fiducial parameters and their adopted fractional uncertainties are:

Parameter	Fiducial	$\delta x/x$	Source
M	$6.5 \times 10^9\, M_\odot$	0.15	EHT + stellar dynamics spread
a_*	0.9	0.11	$a_* = 0.9 \pm 0.1$
$\lambda_{\rm Edd}$	3×10^{-6}	0.5	Bolometric correction uncertainty
κ	0.1	1.0	Factor ~ 3 range: 0.03–0.1

Applying Eq. (19):

$$\frac{\delta P_{\rm BZ}}{P_{\rm BZ}} = \sqrt{0.15^2 + \tfrac{1}{4}(0.5)^2 + \tfrac{1}{4}(1.0)^2 + 4(0.11)^2} = \sqrt{0.023 + 0.063 + 0.250 + 0.048} = 0.62. \tag{20}$$

The error budget is dominated by κ (41% of the variance), followed by $\lambda_{\rm Edd}$ (10%), spin (8%), and mass (4%).

Monte Carlo validation. To test whether the analytic log-space approximation holds (the uncertainties are large enough that higher-order terms may matter), we perform a 5×10^4-draw Monte Carlo simulation in which each parameter is drawn from a log-normal distribution with the above widths. Figure 5 shows the results. The fiducial $\log_{10} P_{\rm BZ} = 44.76$; the Monte Carlo 68% confidence interval is [44.5, 45.0], corresponding to ~ 0.5 dex. The MC distribution is somewhat broader than the analytic estimate (which gives ~ 0.27 dex as $\delta P/P/\ln 10$) because the large $\delta\kappa/\kappa \sim 1$ introduces non-Gaussian tails—the analytic formula, which assumes small fractional errors, underestimates the spread. This confirms that MC propagation is needed for quantities with large ($\gtrsim 0.5$) fractional uncertainties, and that the analytic formulas serve as useful lower bounds.

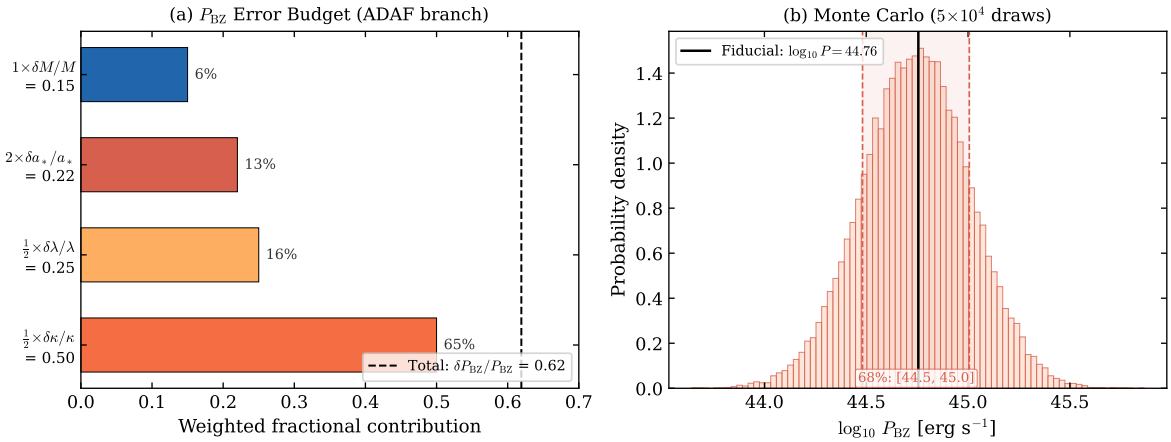

Figure 5: M87* uncertainty propagation (ADAF branch, $\kappa = 0.1$). (a) Error budget: weighted fractional contributions to $\delta P_{\rm BZ}/P_{\rm BZ}$, with the percentage of total variance labeled. The ADAF coupling κ dominates at 41%. (b) Monte Carlo distribution of $\log_{10} P_{\rm BZ}$ from 5×10^4 draws. The 68% CI (dashed red lines) spans [44.5, 45.0], or ~ 0.5 dex. The analytic estimate (~ 0.27 dex) underestimates the spread because $\delta\kappa/\kappa \sim 1$ violates the small-error assumption.

6.3 Implementation status

The analytic uncertainty formulas are fully specified in this paper; the Monte Carlo demonstration is a standalone script. *Automated uncertainty propagation is not yet integrated into* `bhruler.py` *V1*, which computes point estimates only. Adding user-supplied error columns and optional MC output is planned for V2. In the interim, users can apply Eqs. (15)–(19) to their own error budgets, or use the dual-branch output spread as a conservative model-dependent uncertainty bracket.

7 Discussion and Outlook

7.1 What the ruler enables

The Black Hole Ruler's contribution is not any single equation—each component is standard [1, 2, 4, 13]. The value lies in the *integrated, schema-consistent pipeline* that:

- Enforces unit consistency across mass scales via the built-in invariant check (Table 1, Fig. 1).
- Makes model dependence explicit rather than hidden behind a single prescription (Table 3, Fig. 3).
- Provides a common output schema for heterogeneous source catalogs (Appendix A).
- Enables direct cross-scale comparisons (Fig. 4).

Specific use cases include: **Instrument matching**—comparing ISCO frequencies to detector bandpasses (LIGO, LISA, PTA; Fig. 4); **Variability rescaling**—normalizing light-curve timescales to t_g for cross-source comparison (as demonstrated for Sgr A*'s $\sim$ 20-minute flares); **Imaging forecasts**—predicting angular shadow sizes for future EHT targets using r_s and angular-diameter

distance; **Growth archaeology**—tracking BH mass and jet power across redshift with a consistent scaling template.

7.2 Limitations

- The B_H normalization (Eq. 7) is order-of-magnitude; precise field strengths require full GRMHD simulations with source-specific accretion rates, magnetic flux histories, and disk–jet coupling [9, 10].
- The TDE logistic boundary (Eq. 13) is a smooth, tunable parameterization rather than a replacement for detailed relativistic disruption calculations [15, 16]. The spin coefficient (0.6) has not been fitted to those calculations.
- The BZ jet-power scaling uses the leading-order a_*^2 term; higher-order corrections can matter for near-extremal spin [8].
- Uncertainty propagation (analytic and MC) is demonstrated in this paper but not yet automated in the code.
- Retrograde ISCO frequencies are implemented in the code but not exercised in the default pipeline.

7.3 Comparison with existing tools

Several related tools and scaling-relation compilations exist: the `kerrpy` package provides Kerr geodesic calculations; Ghisellini et al. [20] compile blazar jet-power scalings; Stone & Metzger [17] provide detailed TDE rate models; and the EHT Collaboration has published extensive tables of M87* parameters [29, 30]. The Ruler differs from these in its *integration*: it combines core GR scales, dual accretion branches, environmental context, and TDE diagnostics into a single pipeline with a uniform schema. No existing tool, to our knowledge, provides all of these in a consistent, single-file, command-line framework with explicit model-dependence flags.

7.4 Future work

Planned extensions include: (i) full spin posteriors via a hierarchical Bayesian combiner; (ii) expanded environment metrics (gas fraction, nuclear stellar profiles, morphology classes); (iii) calibrated TDE rates with a hierarchical fit to current samples [17–19]; (iv) automated analytic and Monte Carlo error propagation in the code; (v) retrograde ISCO frequency output by default; and (vi) a ~ 50-object catalog with Gold/Silver/Bronze quality grades, advancing toward a community-standard Black Hole Atlas.

Acknowledgements

The author thanks colleagues and collaborators for discussions that informed parts of this framework. This work made use of `numpy` [44] and `matplotlib` [45].

A Output Schema and Column Mapping

Table 5 provides the complete mapping from equations in this paper to output columns in `bhruler.py`. All columns follow a consistent naming convention; accretion-dependent columns carry a suffix (`_eta` for the efficiency bridge, `_adaf` for the ADAF/RIAF branch).

Table 5: Output column schema for `bhruler.py`. "Eq." refers to equations in this paper. Columns marked "opt." require optional inputs (σ, R_e). Accretion columns are duplicated per branch when `--branch both` is selected.

Column	Units	Equation	Description
Core ruler (mass & spin only)			
`t_g_s`	s	Eq. 1	Gravitational time $t_g = GM/c^3$
`r_s_km`	km	Eq. 1	Schwarzschild radius $r_s = 2GM/c^2$
`f_ISCO_Schw_Hz`	Hz	Eq. 2	Schwarzschild ISCO frequency
`r_ISCO_rg`	r_g	Eq. 3	Kerr ISCO radius (prograde)
`f_ISCO_Kerr_Hz`	Hz	Eq. 4	Kerr ISCO frequency (prograde)
`f_ratio`	—	—	$f_{\rm ISCO}^{\rm Kerr}/f_{\rm ISCO}^{\rm Schw}$
Accretion & jets (per branch)			
`B_H_G_eta`	G	Eqs. 5,7	Horizon field (η-bridge)
`P_BZ_erg_s_eta`	$\mathrm{erg\,s^{-1}}$	Eq. 8	BZ jet power (η-bridge)
`B_H_G_adaf`	G	Eqs. 6,7	Horizon field (ADAF)
`P_BZ_erg_s_adaf`	$\mathrm{erg\,s^{-1}}$	Eq. 8	BZ jet power (ADAF)
Environment (opt.)			
`rinfl_pc`	pc	Eq. 9	Sphere of influence
`rinfl_over_Re`	—	—	$r_{\rm infl}/R_e$
TDE diagnostics			
`t_fb_days`	days	Eq. 12	Fallback time (solar-type star)
`tde_possible`	bool	Eq. 13	$S < 0.5$: disruption dominates

B Physical Constants

Table 6: Physical constants used in `bhruler.py` (SI). These are defined at the top of the source file and are trivially replaceable.

Constant	Value	Source
G	$6.67430 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$	CODATA 2018
c	$2.99792458 \times 10^{8}\ \mathrm{m\,s^{-1}}$	Exact (SI definition)
$M_\odot$	1.98847×10^{30} kg	IAU 2015 nominal
1 pc	3.0857×10^{16} m	IAU 2012

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